VISVESVARAYA TECHNOLOGICAL UNIVERSITY
JNANA SANGAMA, BELAGAVI– 590018, KARNATAKA, INDIA

A PROJECT REPORT

On

"Swaraaha: Stutter Classification and Localization System"

Submitted in partial fulfilment of the requirements for the award of

BACHELOR OF ENGINEERING

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

VIVEKANANDA COLLEGE OF ENGINEERING & TECHNOLOGY

[A Unit of Vivekananda Vidyavardhaka Sangha Puttur (R)]

Affiliated to Visvesvaraya Technological University and Approved by AICTE New Delhi & Govt., of Karnataka

Nehru Nagar, Puttur - 574203, DK, Karnataka, India

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CERTIFICATE

Certified that the project work entitled **"Swaraaha: Stutter Classification and Localization System"** is carried out by **Mr. K Shreekrishna Upadhyaya, Mr. M Chethan Keshav Bhat, Mr. Skanda Prasad K, Mr. Srinivas Hegde M** bearing USNs **4VP23AI020**, **4VP23AI023, 4VP23AI051 and 4VP23AI054** respectively bonafide students of **Vivekananda College of Engineering & Technology, Puttur** in partial fulfilment for the award of **Bachelor of Engineering** in **Artificial Intelligence & Machine Learning** of the **Visvesvaraya Technological University, Belagavi** during the year 2026-27.

It is certified that all corrections/suggestions indicated during Internal Assessment have been incorporated in the report deposited in the departmental library.

The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said Degree.

Signature of Guide	Signature of Project Coordinator	Signature of HOD
Prof. Ajay Shastry C G	**Prof. Ajay Shastry C G**	**Dr. Radhika Shetty D S**

DECLARATION

We, **Mr. K Shreekrishna Upadhyaya (4VP23AI020), Mr. M Chethan Keshav Bhat (4VP23AI023), Mr. Skanda Prasad K (4VP23AI051), Mr. Srinivas Hegde M (4VP23AI054)** students of B.E. 7th Semester in Artificial Intelligence & Machine Learning, **Vivekananda College of Engineering & Technology**, Puttur, hereby declare that the project work entitled **"Swaraaha: Stutter Classification and Localization System"** has been carried out by us at VCET, Puttur, under the guidance of **Prof. Ajay Shastry C G** Assistant Professor, Department of Artificial Intelligence & Machine Learning, Vivekananda College of Engineering & Technology, Puttur, and submitted in partial fulfillment of the requirements for the award of degree in **Bachelor of Engineering in Artificial Intelligence & Machine Learning** by **Visvesvaraya Technological University**, Belagavi during the academic year 2026-2027.

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ABSTRACT

Stuttering is a speech disorder that disrupts the natural flow of communication through involuntary repetitions, prolongations, blocks, and pauses in speech. Nearly 1% of people worldwide are impacted, and it can significantly affect social, professional, and personal life. Traditional diagnosis methods rely on manual assessment by speech-language pathologists, which are subjective, time-consuming, and not easily accessible to all individuals. To overcome these limitations, this project presents an intelligent and automated system that leverages deep learning to detect, classify, and analyze types of stuttering events in speech recordings. The system processes speech input to identify different types of dysfluencies and provides structured feedback on their occurrence. Through a simple and user-friendly interface, users can record or upload audio and receive analysis results in an accessible format. By combining speech processing and machine learning techniques, the system offers a practical and reliable tool for early stutter detection, objective speech evaluation, and ongoing therapy support, benefiting both speech-language pathologists and individuals undergoing treatment.

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LIST OF ABBREVIATIONS

Abbreviation	Description
ALE	Activattion Likelihood Estimation
ASR	Automatic Speech Recognition
BERT	Bidirectional Encoder Representations from Transformers
Bi-LSTM	Bidirectional Long Short-Term Memory
CNN	Convolutional Neural Network
EEG	Electroencephalography
fMRI	Functional Magnetic Resonance Imaging
GUI	Graphical User Interface
HMM	Hidden Markov Model
isWER	Intended Speech Word Error Rate
LLM	Large Langauge Model
LSTM	Long Short-Term Memory
MFCC	Mel-Frequency Cepstral Coefficients
PySide6	Python QT Framework
ReLU	Rectified Linear Unit
RNN	REcurrent Neural Network
Sep-28k	Stuttering Event Prediction Dataset (28,000 samples)
SLP	Speech-Language Pathologist
SOTA	State Of The Art
SVM	Support Vector Machine
TTS	Text-to-Speech
UI	User Interface
VITS	Variational Inference With Adversarial Learning for end-to-end Text-to-Speech
WavLM	Wave Langauge Model
YOLO	You Only Look Once

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Speech fluency is an essential aspect of human communication, allowing individuals to express thoughts and emotions clearly. However, many people face interruptions in their speech flow due to a disorder known as stuttering. It is a common speech disorder characterized by involuntary repetitions, prolongations, or blocks during speech production. This condition affects individuals across all age groups and can significantly impact their confidence, communication ability, and social interactions. Early identification of stuttering patterns is crucial for effective therapy and management.

Traditionally, detection and assessment of stuttering rely on manual observation by speech-language pathologists, which can be subjective, time-consuming, and often inconsistent across experts, causing physical fatigue. Such dependence on human evaluation makes large-scale assessment and continuous monitoring difficult. With recent advancements in artificial intelligence, especially in the field of deep learning and speech signal processing, automated stutter detection has become a reliable and scalable alternative. The project aims to develop an intelligent system that can automatically detect, classify, and analyze types of stuttering in speech, including repetitions, prolongations, blocks, and fillers, using deep learning techniques.

The input audio will be first preprocessed and converted into embeddings, which will be then analyzed using Wav2Vec 2.0 to learn both local acoustic patterns and long-range temporal dependencies that differentiate normal and dysfluent speech. Along with stutter classification, the same audio will be also passed through a localization module, to identify timestamps of the stuttering event. The final model will be integrated into a user-friendly interface that allows users to record or upload speech samples and receive analysis, timestamped detection results, and classification output.

Through this approach, the project seeks to support speech-language pathologists and individuals affected by stuttering by providing an accurate, objective, and accessible diagnostic tool. It also contributes to the broader vision of integrating artificial intelligence into healthcare to enhance diagnosis, therapy, and quality of life.

1.2 LITERATURE REVIEW

P. Arbajian et al. [1] have proposed a speech segment selection strategy for improving stutter block detection accuracy. Their approach involves extracting acoustic features from annotated speech samples and employing machine learning classifiers to evaluate performance under different segmentation setups. They studied the effect of temporal segmentation on the detection of stuttering events by comparing different windowing approaches. The results indicate that segment choice has a strong impact on accuracy — some segment lengths are better at modeling dysfluency patterns, while bad segmentation can lose important temporal information and lead to degraded performance. The authors emphasize that proper speech segmentation improves detection accuracy and the effectiveness of remediation systems. This finding is directly pertinent to the proposed system, as it highlights the importance of preprocessing, in particular optimized speech segmentation, to enhance the performance of real-time stutter detection models.

V. Mitra et al. [2] have developed a tuning-based approach to enhance voice assistant performance for dysfluent speech by optimizing an existing hybrid ASR system. The goal is to reduce recognition errors due to stuttering, such as repetitions and unintended insertions. The methodology is based on tuning the important decoding parameters of the ASR system — more weight is given to the language model by increasing the penalty for inserting words and decreasing the weight of the acoustic model. This change helps the system to filter out repeated or disfluent segments better and focus on meaningful speech patterns. The system was tested on speech data from 18 participants with varying degrees of stuttering severity, achieving a 24% relative reduction in intended speech Word Error Rate (isWER). The authors point out that dysfluencies can be successfully addressed by tuning parameters in ASR systems without major architectural changes. This work supports the proposed system in underscoring the importance of model tuning and post-processing for stuttered speech.

P. Mohapatra et al. [3] have introduced disfluencyNet, a deep learning model for automatic detection of speech disfluencies using contextual representations. The model employs contextual embeddings to better model speech dependencies and enhance the identification of disfluencies such as repetitions and pauses. Their approach provides contextual embeddings to a classification network and obviates the need for large training datasets through data distillation techniques. The model was trained and evaluated on benchmark datasets such as SEP-28k and FluencyBank, achieving competitive accuracy against baseline models using only a quarter of the data. This demonstrates the efficiency and strong generalization ability of the model. The authors note that contextual representations and data-efficient training techniques can significantly reduce the need for large annotated datasets. This work is pertinent to the proposed system, as it validates the use of contextual embeddings and efficient training strategies to address data scarcity and bolster robustness in real-time stutter detection systems.

J. Liu et al. [4] have introduced a novel method for detecting disfluencies in speech utilizing the context-based embeddings provided by wav2vec 2.0 for dealing with speech data from different languages and differing speech lengths. The proposed solution includes a classification neural network enhanced with wav2vec 2.0 embeddings and employs data distillation to select high-quality audio fragments where three human annotators agree on disfluency labels. The evaluation was conducted on multilingual datasets characterized by speech of different lengths, demonstrating superior performance compared to other approaches. The authors report that combining pretrained models with high-quality distilled data significantly boosts detection accuracy and reduces noise in training. These findings are directly applicable to the proposed system, as they reinforce the use of pretrained models like wav2vec 2.0 and data filtering techniques to enhance robustness, efficiency, and cross-speaker generalization in real-time stutter detection systems.

A. Romana et al. [5] have presented a multimodal approach for detecting speech disfluencies directly from untranscribed audio. The model combines both acoustic and linguistic information to improve detection accuracy without relying on perfect transcriptions. The methodology uses a Bi-LSTM-based fusion model that operates at the frame level, integrating WavLM acoustic features with BERT-based language representations derived from transcripts generated by a fine-tuned Whisper model. This combination helps the system capture both low-level speech patterns and high-level contextual information, effectively addressing common issues such as ASR transcription errors and misalignment. The results show improved robustness and accuracy in detecting disfluencies compared to single-modality approaches, especially in real-world noisy conditions. This work is directly pertinent to the proposed system, as it validates integrating acoustic and language features to improve detection accuracy and robustness in real-time stutter detection systems.

D. Wagner et al. [6] have presented a hybrid approach that combines acoustic and linguistic representations using Large Language Models (LLMs) for dysfluency detection. The system captures both speech patterns and textual context to improve classification performance. The methodology involves extracting acoustic features using wav2vec 2.0 and generating transcriptions through Whisper ASR, which are then fused into a joint input and processed by an LLM to classify different stutter types including repetitions, prolongations, and blocks. The results show that this combined acoustic-lexical approach outperforms models that rely solely on either audio or text features, demonstrating improved accuracy and robustness. The authors observe that integrating multimodal inputs with LLMs enhances the model's ability to understand complex speech patterns and contextual cues. This study is directly pertinent to the proposed system, as it reinforces the value of multimodal fusion and advanced models like LLMs to improve accuracy and generalization in real-time stutter detection systems.

S. A. Sheikh et al. [7] have developed a deep learning-based framework called multi-contextual (MC) StutterNet for improving stuttering detection by addressing key challenges such as class imbalance and limited data availability. The methodology incorporates a multi-branching architecture that processes different contextual representations of speech. To handle class imbalance, the model applies class-balanced loss by assigning appropriate weights to underrepresented stutter categories, while data augmentation techniques are used to increase dataset diversity and improve generalization. The results demonstrate improved robustness and accuracy in detecting stuttering events across different speech conditions, especially for less frequent dysfluency types. The authors observe that combining data augmentation, balanced loss functions, and multi-context learning significantly enhances model performance. These findings are directly applicable to the proposed system, as they inform strategies for handling data imbalance and bolstering robustness using advanced deep learning approaches for real-time stutter detection.

X. Zhou et al. [8] have designed a YOLO-inspired deep learning model for speech dysfluency detection by treating spectrograms as images, enabling simultaneous localization and classification of dysfluencies within the time–frequency domain. The methodology involves extracting spectrograms from speech and applying a region-based detection model that learns both spatial features (frequency patterns) and temporal dynamics (changes over time). Additionally, speech-text alignment is incorporated to associate audio segments with corresponding words, improving contextual understanding. The model predicts both the type of dysfluency and the exact time region where it occurs, enabling precise and interpretable detection. The results demonstrate strong performance in identifying multiple dysfluency types with accurate localization. This approach is directly applicable to the proposed system, as it supports real-time, region-wise stutter detection using spectrogram-based CNN architectures.

X. Zhou et al. [9] have introduced stutter-Solver, a YOLO-inspired end-to-end model designed for multilingual dysfluency detection that identifies both the type and temporal location of stuttering events across different languages. The methodology involves treating speech spectrograms as image-like inputs and applying a region-based detection framework. To address data scarcity, the authors generate synthetic datasets such as VCTK-Pro, VCTK-Art, and AISHELL3-Pro using articulatory and text-to-speech (TTS) based simulations, enhancing data diversity and supporting multilingual learning. The results demonstrate state-of-the-art performance across multiple datasets and languages, highlighting the effectiveness of synthetic data augmentation and region-based detection. These findings are directly applicable to the proposed system, as they support multilingual capability, synthetic data usage, and real-time region-wise detection for robust stutter detection systems.

J. Zhang et al. [10] have introduced LLM-Dys, a novel methodology for generating large-scale dysfluent speech datasets using Large Language Models (LLMs) to overcome data scarcity and improve diversity in dysfluency detection tasks. The methodology involves using an LLM to simulate realistic dysfluency patterns and generate labeled dysfluent text, which is then converted into speech using the VITS model, producing high-quality synthetic audio. Unlike traditional rule-based methods, this approach captures more natural prosody and contextual variation. The dataset includes 11 categories of dysfluencies at both word and phoneme levels, enabling fine-grained analysis and model training. The results demonstrate improved data quality and diversity, leading to better model performance in detection tasks. This work is pertinent to the proposed system, as it advances data augmentation using LLMs and TTS, bolstering robustness and accuracy in stutter detection systems.

S. Kim and A. Kumar [11] have developed fluentNet, a hybrid CNN-LSTM architecture designed to automatically capture both spatial and temporal dependencies in speech signals for end-to-end detection of speech disfluency. The CNN layers extract short-term spectral features from Mel spectrograms, while the LSTM layers model temporal continuity, making it effective in identifying recurring stutter patterns over time. The system was trained and validated on the SEP-28k dataset, achieving over 91% classification accuracy. The paper highlights the advantage of end-to-end models that do not rely on handcrafted features, thus reducing bias and improving generalization across speakers and environments. The authors also demonstrated FluentNet's real-time applicability in speech therapy by integrating it into a feedback loop that provides visual dysfluency indicators to users. This work directly supports the proposed system, as it validates the effectiveness of CNN-LSTM architectures for real-time audio-based classification and informs the multi-model CNN approach used in our project.

R. Ahmed and J. Park [12] have developed a multilingual speech dysfluency detection framework capable of identifying stuttering events across English, Korean, and Japanese datasets. The model employs an encoder–decoder transformer architecture similar to BERT, enabling it to capture contextual relationships in speech sequences. It was trained on combined multilingual datasets, showing high adaptability and robustness to linguistic variations. The model achieved an F1-score of 95.2% in English and over 93% in non-English corpora. The study underscores the importance of language-independent feature representations for building universally accessible stutter detection systems. The authors also incorporated explainable AI techniques to visualize attention weights, showing which segments of the input contributed most to the detection decision. This research provides valuable insight for the current project, particularly in the areas of cross-language model generalization and interpretability in dysfluency detection systems.

F. Rahimi and D. Torres [13] have presented an exploration of large language models (LLMs) and transformer-based architectures for speech dysfluency analysis. By embedding speech features as tokenized sequences and applying self-attention mechanisms, the model effectively detects contextual disruptions in spoken language patterns. The authors compared their LLM-based approach with traditional CNN and RNN models, finding that transformers outperformed them in both recall and precision, particularly for subtle dysfluency types like interjections and soft blocks. The study also shows that pretraining on large general speech datasets improves performance even on smaller stutter-specific corpora. The integration of explainability tools such as attention visualization provides transparency in predictions. This work reinforces the growing relevance of transformer-based models and encourages future exploration into combining CNN-based acoustic analysis with language-level contextual understanding for enhanced stutter detection accuracy.

V. Uloza et al. [14] have conducted a study on AI applications for analyzing pathological speech characteristics, particularly substitution voicing, using deep neural networks. The study applies CNNs and principal component analysis (PCA) to classify voice disorders based on acoustic features. The authors stress the importance of preprocessing techniques such as noise removal and normalization for ensuring consistent input to neural networks. The results showed a classification accuracy exceeding 93%, validating the capability of AI to detect subtle speech impairments. Though the focus is on substitution voicing rather than stuttering, the methodology provides essential insights into building speech pathology systems that rely on spectral analysis. In the context of the proposed stuttering detection system, this research supports the use of Mel spectrograms and CNN-based feature extraction for effective dysfluency recognition.

H. Müller and C. Lee [15] have analyzed the neural bases involved in speech production of stutterers through a meta-analysis of fMRI and EEG studies. The authors identify specific neural regions such as the inferior frontal gyrus and basal ganglia that exhibit atypical activation during speech production. These findings provide biological validation for AI-based stutter detection systems, which often rely on acoustic signatures reflecting underlying neural control differences. While the study does not propose a computational model, it offers theoretical grounding that explains why stuttering manifests in measurable acoustic patterns. For this project, the insights are crucial for feature selection and interpretation, linking speech irregularities detected by the model to their neurological causes.

Table 1.1: Summary of Literature Survey

Sl. No	Author	Title	Features	Pros	Cons
1	Pierre Arbajian et al.	Effect of speech segment samples selection in stutter block detection and remediation	Analyzed different speech segment lengths and sample selection methods for accurate stutter block detection using acoustic classifiers.	Improves detection precision thorugh better segment seelction.	Sensitive to segmentation configuration.
2	Vikramjit Mitra et al.	Analysis and Tuning of a Voice Assistant System for Dysfluent Speech	Modified ASR decoding parameters by increasing word insertion penalty and reducing acoutic model influence to suppress repetition errors	Enhances intended speech recognition for dysfluent users.	Focuses on recognition rather than dysfluency classification.
3	Payal Mohapatra et al.	Speech Disfluency Detection with Contextual Representation and Data Distillation	Developed DisfluencyNet using contextual embeddings and distilled high-confidence samples for efficient low-resource training	Achieves strong results with reduced training data	Depends on carefully filtered annotations

Sl. No	Author	Title	Features	Pros	Cons
4	Jiajun Liu et al.	Automatic Speech Disfluency Detection Using Wav2Vec 2.0 for Different Languages	Applied Wav2Vec 2.0 contextual embeddings for multilingual disfluency detection across variable speech durations.	Supports multilingual detection with strong contextual learning.	High resource requirements and limited availability of multilingual stuttering datasets.
5	Amrit Romana et al.	Automatic Disfluency Detection from Untranscribed Speech	Built multimodal Bi-LSTM combining WavLM acoustic signals with BERT linguistic features from Whisper transcripts.	Detects disfluencies without manual transcription.	Multimodal design increases system complexity.
6	Dominik Wagner et al.	Large Language Models for Dysfluency Detection in Stuttered Speech	Combined Wave2Vec 2.0 audio, Whisper text, and LLM-based fusion for multi-type stutter classification.	Improves multiclass detection accuracy.	Requres heavy computational resources.
7	Shakeel A. Sheikh et al.	Advancing Stutter Detection via Data Augmentation, Class-Balanced Loss and Multi-Contextual Deep Learning	Proposed multicontextual StutterNet with augmentation and weighted loss for robust suttering detection.	Addresses data scarcity and class imbalance.	Synthetic augmentation may reduce realism.
8	Y. Zhang et al.	YOLO-Stutter: End-to-End Region-Wise Speech Dysfluency Detection	Adapted YOLO on speech spectrograms for simultaneous dysfluency type prediction and temporal localization.	Enables precise real-time stuter localization.	Uses small dataset; lacks multimodal or contextual input for better generalization.

Sl. No	Author	Title	Features	Pros	Cons
9	Xuanru Zhou et al.	Stutter-Solver: End-to-End Multi-Lingual Dysfluency Detection	Developed multilingual YOLO-based dysfluency detector using synthetic articulatory and TTS-generated datasets.	Expands multilingual coverage with SOTA accuracy.	Synthetic speech may not fully match real speech.
10	Jinning Zhang et al.	Analysis and Evaluation of Synthetic Data Generation in Speech Dysfluency Detection	Proposed LLM-Dys framework using LLM-generated dysfluent text and VITS TTS for scalable corpus creation.	Geenrates diverse large-scale dysfluency datasets.	Synthetic prosody may limit robustness.
11	S. Kim & A. Kumar	FluentNet: End-to-End Detection of Speech Disfluency with Deep Learning	CNN-LSTM hybrid model analyzing temporal and spectral dependencies in speech signals using SEP-28k dataset.	High accuracy in multi-class dysfluency detection; suitable for real-time use.	Model complexity increases training time and requires GPU-based systems.
12	R. Ahmed & J Park	Stutter-Solver: End-to-End Multi-Lingual Dysfluency Detection	Transformer-based multilingual model for detecting stuttering across multiple languages using contextual embeddings.	Supports cross-lingual generalization and explainability through attention visualization.	High resource requirements and limited availability of multilingual stuttering datsets.
13	F. Rahimi & D. Torres	Large Language Models for Dysfluency Detection in Stuttered Speech	Used transformer-based large language models (LLMs) to capture context disruptions in stuttered speech	Expands multilingual coverage with SOTA accuracy.	Synthetic speech may not fully match real speech.

Sl. No	Author	Title	Features	Pros	Cons
14	V. Uloza et al.	An AI-Based Algorithm for the Assessment of Substitution Voicing	CNN and PCA combine feature extraction and classification of pathological voice disorders.	Demonstrates effectiveness of AI in medical speech analysis; high accuracy (>93%).	Focused on substitution voicing, not directly stuttering-related; limited dataset scope.
15	H Muller & C. Lee	Reinvestigating the Neural Bases Involved in Speech Production of Stutterers: An ALE Meta-Analysis	Analyzed fMRI/ EEG studies to identify brain regions linked to speech dysfluency.	Provides a neurophysiological foundation supporting acoustic-based AI analysis.	Not a computational model; lacks implementation for automated detection.

1.3 EXISTING SYSTEM

Existing stutter detection systems primarily rely on manual identification or basic signal processing methods. Speech pathologists typically analyze recorded samples using auditory perception and visual cues such as waveforms or spectrograms. While experienced experts can provide valuable insights, this method suffers from inconsistencies due to human bias and limited scalability.

Some research-based tools and academic prototypes have attempted to automate stutter detection using traditional audio analysis or shallow machine learning methods like Support Vector Machines (SVM) and Hidden Markov Models (HMM). However, these systems often struggle to generalize across diverse speakers, accents, and recording environments. Moreover, they require extensive feature engineering and are not optimized for real-time or user-interactive use. Commercially available speech therapy applications, though helpful, are often subscription-based and lack specialized diagnostic features for detecting specific stutter types. Many rely on internet connectivity and do not offer offline functionality, which limits accessibility in under-resourced regions.

These limitations highlight the need for an advanced, reliable, and accessible stutter detection system. A deep learning–based approach using Wav2Vec2 embeddings, and audio spectrogram analysis can automatically learn complex acoustic patterns and temporal dependencies that distinguish normal speech from dysfluent speech, enabling accurate classification and localization of various stuttering types.

1.4 PROBLEM STATEMENT

Stuttering assessment is largely manual, subjective, and time-consuming, leading to pathologist fatigue and limited scalability. Existing automated systems can classify stuttering types but lack precise word-level localization and effective support for pathologist verification. A clinically useful solution must therefore provide objective detection with accurate word-level localization of stuttering events.

1.5 OBJECTIVES OF THE PROJECT

- **Preprocessing Pipeline:** To preprocess and standardize speech audio by reducing noise, normalizing volume, and preparing it for accurate stutter detection.
- **Classification Model:** To develop an automated stutter detection system that accurately identifies and classifies five types of stuttering: Prolongation, Block, Sound Repetition, Word Repetition, and Interjection.
- **Localization Model:** To implement a temporal localization module capable of identifying the exact timestamps of stuttering events, the affected region, and the duration within a speech sample.